

Case Study 2: Mutual Fund Performance Analytics (Medium Challenge)

Difficulty: (Medium)

Estimated Time: 90–120 minutes

Skills Covered:

- Python Functions
- File Handling
- Exception Handling
- NumPy
- Pandas
- Data Cleaning
- GroupBy
- Aggregation
- Merge
- Basic Financial Calculations

Business Scenario

ABC Asset Management Company (AMC) manages multiple mutual funds across India.

The management wants a Python application that analyzes fund performance, investor portfolios, and investment returns to help recommend the best-performing funds.

The company provides the following datasets.

funds.csv

investors.csv

transactions.csv

nav_history.csv

Part 1 - Read Data

Read all CSV files using Pandas.

funds.csv

investors.csv

transactions.csv

nav_history.csv

Part 2 - Data Cleaning

Perform the following:

Remove duplicate rows.

Check missing values.

Fill missing NAV using Forward Fill.

Replace missing InvestorType with "Retail".

Remove rows having negative NAV.

Convert Date columns into datetime format.

Part 3 - Merge Data

Merge all four datasets into one DataFrame.

Required columns

Investor Name

Fund Name

Category

AMC

State

Units Purchased

Purchase NAV

Latest NAV

Part 4 - Create New Columns

Calculate

Investment Amount

Investment Amount = Units Purchased × Purchase NAV

Current Value

Current Value = Units Purchased × Latest NAV

Profit

Profit = Current Value – Investment Amount

ROI % = $(\text{Current Value} - \text{Investment Amount}) / \text{Investment Amount} \times 100$

Part 5 - NumPy Tasks

Using NumPy calculate

Average NAV

Maximum NAV

Minimum NAV

Variance of NAV

Standard Deviation of NAV

Rolling Average (window = 5)

Part 6 - Pandas Analysis

Display

Top 5 investors based on investment amount

Top 5 profitable funds

Worst performing fund

Highest NAV fund

Lowest NAV fund

Part 7 - GroupBy

Group by Category

Find

Average ROI

Average NAV

Total Investment

Group by AMC

Find

Number of Funds

Average NAV

Total Investment

Group by State

Find

Number of Investors

Total Investment

Average ROI

Group by Investor Type

Find

Total Investment

Average Profit

Part 8 - Detect Issues

Find

Duplicate NAV records

Negative NAV

Future dates

Missing Fund IDs

Missing Investor IDs

Invalid Purchase NAV (<0)

Part 9 - Finance Metrics

Calculate

ROI = (Current Value - Investment Amount) / Investment Amount × 100

Absolute Return

Current Value - Investment Amount

Annual Return

Assume holding period is 1 year.

Volatility

Using

np.std(NAV)

Sharpe Ratio

Assume

Risk Free Rate = 6%

Formula

Sharpe Ratio = (Return-Risk Free Rate) / Volatility

Part 10 - Export Reports

Generate

TopFunds.xlsx

InvestorSummary.xlsx

CategorySummary.csv

Expected Outputs

Top Performing Funds

Highest ROI

Highest Profit

Highest NAV

Worst Performing Fund

Lowest ROI

State-wise Investment

AMC-wise Investment

Category-wise ROI
